CHOWLA'S COSINE PROBLEM WITH NEAR-UNIT COEFFICIENTS

KNOWLEDGE SENSE

ABSTRACT. Let $a_1, \dots, a_N$ be distinct positive integers and let $s_1, \dots, s_N$ be real numbers in the interval $[0.99, 1.01]$. We prove that

$$\min_{x \in \mathbb{R}} \sum_{j=1}^{N} s_j \cos(2\pi a_j x) \le -\frac{N^{1/9}}{20}.$$

Bedert recalled a question of Ruzsa asking how negative such cosine sums must be; the lower bound recorded there is of order $\log N$. Our proof adapts a Fourier-analytic construction from Bedert's work on Chowla's cosine problem, where all coefficients are equal to one. Allowing the coefficients to vary creates an error term for each translation; after summation over all translations, this error is controlled by a counting identity of Shankar. The required bounds for two quadratic forms in five variables, each equal to 0 or lying in $[0.99, 1.01]$, reduce to finitely many exact checks, which are listed in an appendix.

1. INTRODUCTION

For a set S of N positive integers, the cosine sum $\sum_{a \in S} \cos(2\pi a x)$ has mean zero over a period and therefore takes negative values. Chowla's cosine problem [3] asks how negative it must become, that is, for the order of growth of

$$K(N) = \inf_{S \subset \mathbb{N},\ |S| = N} \Big(- \min_{x \in \mathbb{R}} \sum_{a \in S} \cos(2\pi a x) \Big).$$

Difference sets of Sidon sets give $K(N) \ll \sqrt{N}$, and Chowla conjectured that this is the correct order; see [1, Section 1]. The resolution of the Littlewood L^1 conjecture by McGehee, Pigno and Smith [6] and by Konyagin [5] gives $K(N) \gg \log N$. Bourgain [2] was the first to go beyond logarithmic bounds, and Ruzsa [8] refined his method to obtain $K(N) \ge \exp(c\sqrt{\log N})$. Polynomial lower bounds were obtained independently by Bedert [1] and by Jin, Milojević, Tomon and Zhang [4]. The third version of [1] gives $K(N) \ge N^{1/5 - o(1)}$, and Shankar [9] removed the subpolynomial factor, proving $K(N) \gg N^{1/5}$.

This paper concerns coefficients that are close to one but are not required to be equal to one. Let $I = [0.99, 1.01]$ and

$$K_I(N) = \inf \Big(- \min_{x \in \mathbb{R}} \sum_{j=1}^{N} s_j \cos(2\pi a_j x) \Big),$$

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where the infimum is taken over all distinct positive integers $a_1, \dots, a_N$ and all $s_1, \dots, s_N \in I$. Bedert [1, Theorems 1.2 and 6.1] proved polynomial lower bounds when the coefficients range over a fixed finite set of nonzero reals; both the exponent and the constant depend on that set. He also observed that the other methods giving superlogarithmic bounds for $K(N)$ depend heavily on the coefficients being equal to one, and he recalled a question of Ruzsa: to estimate $K_I(N)$ [1, Section 1]. The bound recorded there is $K_I(N) \gg \log N$, which follows from the L^1 theorem of [6]. Our main result is a polynomial lower bound.

Theorem 1.1. *Let $N \geq 1$, let $a_1, \dots, a_N$ be distinct positive integers, and let $s_1, \dots, s_N \in [0.99, 1.01]$ be real numbers. Then*

$$\min_{x \in \mathbb{R}} \sum_{j=1}^{N} s_j \cos(2\pi a_j x) \leq -\frac{N^{1/9}}{20}.$$

Equivalently, $K_I(N) \geq N^{1/9}/20$ for every $N \geq 1$.

There is no restriction on the size of the frequencies, on their gaps, or on the number of distinct coefficient values. Since all coefficients may equal one, $K_I(N) \leq K(N) \ll \sqrt{N}$; thus $K_I(N)$ has polynomial order of growth (see Section 6.1). We do not know the correct exponent.

Theorem 1.1 does not follow from the case of unit coefficients by a direct perturbation. Replacing every s_j by 1 changes the sum by up to $0.01N$ in the supremum norm, and for large N this exceeds the quantity being estimated, since $K_I(N) \ll \sqrt{N}$. Bedert's argument for a finite set of coefficients separates the coefficient classes by convolution powers and the inversion of a Vandermonde matrix [1, Section 6], and its constants depend on the set of coefficients; it gives no uniform statement for a continuum of coefficient values.

Throughout, $e(x) = e^{2\pi i x}$. As in [1] and [9], we work with a symmetric formulation.

Theorem 1.2. *Let $A = -A$ be a nonempty finite subset of $\mathbb{Z} \setminus \{0\}$, let $n = |A|$, and let $s \colon A \to [0.99, 1.01]$ satisfy $s_{-a} = s_a$ for all $a \in A$. Put*

$$F(x) = \sum_{a \in A} s_a \, e(ax) \qquad \text{and} \qquad K = -\min_{x \in \mathbb{R}} F(x).$$

Then $n \leq 10^8 K^9$.

Outline. For $t \in \mathbb{Z}$ let

$$B_t = \{m \in A : m - t \in A,\ m + t \notin A\}, \qquad C_t = \{m \in A : m - t \in A,\ m + t \in A\}.$$

Thus $A \cap (A + t)$ is the disjoint union of B_t and C_t, and B_t is the asymmetric part of $A \cap (A + t)$ studied in [1, Section 7] and [9]. The proof has three steps.

First, a Roth-type argument gives $\sum_{t \in A} |A \cap (A + t)| \geq n^2/(2K)$ when $n \geq 3K^2$ (Lemma 2.2). Shankar's identity [9, Lemma 3.1] shows that at least a third of this sum comes from the sets B_t (Lemma 2.3), and from it we derive $\sum_{t \in A} |C_t| \leq 2 \sum_{t \in A} |B_t|$ (Corollary 2.4).

Second, for each $t \neq 0$ Bedert [1, Section 7] constructs two real trigonometric polynomials ψ_t and ϕ_t with $|\phi_t| \leq \psi_t + O(K^2)$, whose Fourier coefficients at m depend only on the coefficients of F at $m - 2t, \dots, m + 2t$. For unit coefficients, the coefficients of ϕ_t on B_t have modulus strictly larger than the largest Fourier coefficient of ψ_t [1, Lemma 7.2]. For coefficients in I these Fourier coefficients are

values of two quadratic forms, and Proposition 3.2 shows that a strict separation persists for all admissible coefficient values.

Third, we test the inequality $|\phi_t| \le \psi_t + O(K^2)$ against $g_t * g_t$, where

$$g_t = F * \big(e(t\cdot)F\big) - F * \big(e(t\cdot)F\big) * \big(e(-t\cdot)F\big).$$

With s extended by zero outside A, the Fourier coefficient of g_t at m is $s_m s_{m-t}(1 - s_{m+t})$. For unit coefficients this is the indicator function of B_t. In general it is bounded below on B_t, vanishes outside $B_t \cup C_t$, and is small on C_t (Lemma 4.1). For a single t the set C_t may be much larger than B_t, so this error cannot be controlled one translation at a time (Remark 4.3). After summation over all $t \in A$, the inequality $\sum_t |C_t| \le 2\sum_t |B_t|$ absorbs the total error with a positive margin (Section 5).

Dependencies. The functions f_t, r_t, ψ_t, ϕ_t and the formula for their Fourier coefficients are taken from [1, Section 7, (36)–(41)]. Lemma 2.2 is Roth's argument [7], in the form of [1, Lemma 4.1], with weights. Lemma 2.3 is [9, Lemma 3.1]; its proof uses only the support of F, and we include it for completeness. The new ingredients are the bounds for the perturbed local quadratic forms in Proposition 3.2, the test function g_t of Lemma 4.1, and the absorption of its error after summation over translations in Section 5. To our knowledge, no polynomial lower bound for $K_I(N)$ has been recorded before.

The exponent $1/9$ is weaker than the exponent $1/5$ available for unit coefficients. The loss comes from the crude bound $\|g_t\|_1 \le 13K^3$ in Lemma 4.1; for unit coefficients Bedert's decomposition [1, Lemma 7.1] controls the corresponding test function much more efficiently. We have not attempted to adapt that decomposition.

Organization. Section 2 collects notation and the two counting inputs. Section 3 recalls Bedert's inequality and states the uniform bounds for the perturbed local quadratic forms. Section 4 introduces the test function g_t and proves an inequality for a single translation. Section 5 sums these inequalities and proves Theorems 1.1 and 1.2. Section 6 contains remarks on upper bounds and constants. Appendix A proves the uniform bounds of Proposition 3.2 by a finite computation with exact numbers.

2. Preliminaries

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ with Haar probability measure. For $g \in L^1(\mathbb{T})$ and $m \in \mathbb{Z}$ put $\widehat{g}(m) = \int_{\mathbb{T}} g(x)e(-mx)\,dx$, and let $(g*h)(x) = \int_{\mathbb{T}} g(x-y)h(y)\,dy$. Then $\widehat{g*h} = \widehat{g}\,\widehat{h}$, Parseval's identity reads $\int_{\mathbb{T}} g\overline{h} = \sum_m \widehat{g}(m)\overline{\widehat{h}(m)}$ for $g, h \in L^2(\mathbb{T})$, and Young's inequality gives $\|g * h\|_1 \le \|g\|_1\|h\|_1$.

Throughout Sections 2 to 5 we fix A, n, s, F and K as in Theorem 1.2, and we put

$$\alpha = \frac{99}{100}, \qquad \beta = \frac{101}{100}, \qquad \eta = \frac{\beta^2}{100} = \frac{10201}{10^6}.$$

We extend s to $\mathbb{Z}$ by $s_m = 0$ for $m \notin A$. Then $\widehat{F}(m) = s_m$ for all m, $s_0 = 0$, $s_{-m} = s_m$, and F is real-valued.

Lemma 2.1. *We have $K \ge \alpha$ and $\|F\|_1 \le 2K$.*

Proof. The function $P = F + K$ is nonnegative and $\int_{\mathbb{T}} P = K$, since $s_0 = 0$. For $a \in A$,

$$\alpha \le s_a = \Big| \int_{\mathbb{T}} P(x) e(-ax)\, dx \Big| \le \int_{\mathbb{T}} P = K.$$

Since $F \ge -K$, we have $|F| \le F + 2K$ pointwise, and integration gives $\|F\|_1 \le 2K$. □

Let

$$T_A = \#\{(m,t) \in A^2 : m - t \in A\}.$$

Lemma 2.2. *If $n \ge 3K^2$, then $T_A \ge n^2/(2K)$.*

Proof. This is Roth's argument [7] as presented in [1, Lemma 4.1], with weights. Let $E(x) = \sum_{a \in A} e(ax)$, which is real-valued because $A = -A$, and let $P = F + K \ge 0$. By Parseval's identity and the Cauchy–Schwarz inequality with respect to the measure $P(x)\,dx$,

$$\alpha^2 n^2 \le \Big(\sum_{a \in A} s_a \Big)^2 = \Big(\int_{\mathbb{T}} PE \Big)^2 \le \int_{\mathbb{T}} P \cdot \int_{\mathbb{T}} PE^2 = K \int_{\mathbb{T}} PE^2.$$

Now $E^2 = \sum_{m,t \in A} e((m-t)x)$, so

$$\int_{\mathbb{T}} PE^2 = \sum_{m,t \in A} \widehat{P}(t-m) = Kn + \sum_{\substack{m,t \in A \\ m-t \in A}} s_{m-t} \le Kn + \beta T_A,$$

where we used $\widehat{P}(0) = K$, $\widehat{P}(k) = s_k$ for $k \ne 0$, and $s_{-k} = s_k$. Hence $\alpha^2 n^2 \le K^2 n + \beta K T_A$. If $n \ge 3K^2$, then $K^2 n \le n^2/3$, and therefore

$$T_A \ge \frac{\alpha^2 - 1/3}{\beta} \cdot \frac{n^2}{K} \ge \frac{n^2}{2K},$$

because $(\alpha^2 - 1/3)/\beta = 19403/30300 > 1/2$. □

For $t \in \mathbb{Z}$ let B_t and C_t be as in the introduction, and put

$$B_{\mathrm{tot}} = \sum_{t \in A} |B_t|, \qquad C_{\mathrm{tot}} = \sum_{t \in A} |C_t|.$$

For fixed t, the sets B_t and C_t partition $\{m \in A : m - t \in A\}$. Summing over $t \in A$ gives

$$T_A = B_{\mathrm{tot}} + C_{\mathrm{tot}}. \tag{2.1}$$

Since $A = -A$, we have $-(A \cap (A+t)) = A \cap (A-t)$, so

$$B_t = \big(A \cap (A+t)\big) \setminus \big(-(A \cap (A+t))\big);$$

this is the set denoted B_t in [1] and [9].

Lemma 2.3 (Shankar [9, Lemma 3.1])**.** *We have $B_{\mathrm{tot}} \ge T_A/3$.*

Proof. We recall the argument of [9] in a slightly rearranged form. Let $P_+ = A \cap \mathbb{N}$, and for $u, v \in P_+$ put $d_{u,v} = 1_A(u-v)$ and $p_{u,v} = 1_A(u+v)$. Every pair $(m,t) \in A^2$ lies in exactly one of the classes

$$\{(u,v),\ (-u,-v),\ (u,-v),\ (-u,v)\}, \qquad u, v \in P_+.$$

Since $A = -A$, on these four pairs the pair $(1_A(m-t), 1_A(m+t))$ equals $(d_{u,v}, p_{u,v})$, $(d_{u,v}, p_{u,v})$, $(p_{u,v}, d_{u,v})$ and $(p_{u,v}, d_{u,v})$, respectively. As $d(1-p)+p(1-d) = d+p-2dp$ for $d, p \in \{0,1\}$, we obtain

$$T_A = 2\sum_{u,v\in P_+}(d_{u,v}+p_{u,v}), \qquad B_{\mathrm{tot}} = 2\sum_{u,v\in P_+}(d_{u,v}+p_{u,v}-2d_{u,v}p_{u,v}).$$

Let $Q = \sum_{u,v} p_{u,v}$, $D = \sum_{u,v} d_{u,v}$ and $J_0 = \sum_{u,v} d_{u,v}p_{u,v}$, all sums over $u, v \in P_+$. Clearly $J_0 \le Q$. Since $0 \notin A$, every pair counted by D has $u \neq v$; since $u - v \in A \cap \mathbb{N}$ when $u > v$, the map $(u,v) \mapsto (v, u-v)$ is a bijection from the pairs counted by D with $u > v$ onto the pairs counted by Q, and the pairs with $u < v$ are handled by exchanging u and v. Hence $D = 2Q$, and so

$$T_A = 6Q, \qquad B_{\mathrm{tot}} = 6Q - 4J_0 \ge 2Q = \frac{T_A}{3}. \qquad \square$$

The following consequences are derived here from Lemma 2.3; the first is the form in which Shankar's identity is used in Section 5.

Corollary 2.4. *We have $C_{\mathrm{tot}} \le 2B_{\mathrm{tot}}$. If moreover $n \ge 3K^2$, then $B_{\mathrm{tot}} \ge n^2/(6K)$.*

Proof. By (2.1) and Lemma 2.3, $C_{\mathrm{tot}} = T_A - B_{\mathrm{tot}} \le 3B_{\mathrm{tot}} - B_{\mathrm{tot}} = 2B_{\mathrm{tot}}$. The second assertion follows from Lemmas 2.2 and 2.3. $\square$

3. Bedert's inequality with perturbed coefficients

Fix $t \in \mathbb{Z} \setminus \{0\}$. Following [1, Section 7], let

$$f_t(x) = 2\big(1+\sin(2\pi t x)\big)F(x).$$

Since $2\sin(2\pi t x) = -i\,e(tx) + i\,e(-tx)$, its Fourier coefficients are

$$\lambda_m := \widehat{f_t}(m) = 2s_m - i\,s_{m-t} + i\,s_{m+t} \qquad (m \in \mathbb{Z}). \tag{3.1}$$

Let

$$r_t(x) = \Big(1 - \cos\big(2\pi t x + \tfrac{\pi}{4}\big)\Big)(f_t * f_t)(x)$$

and

$$\psi_t(x) = \frac{r_t(x)+r_t(-x)}{2}, \qquad \phi_t(x) = \frac{r_t(x)-r_t(-x)}{2},$$

and let $\omega = \frac{1}{2}\exp(i\pi/4) = (1+i)/(2\sqrt{2})$.

Lemma 3.1 (Bedert [1, Section 7])**.** *The functions ψ_t and ϕ_t are real trigonometric polynomials, ψ_t is even, ϕ_t is odd, and*

$$|\phi_t(x)| \le \psi_t(x) + 32K^2 \qquad (x \in \mathbb{T}). \tag{3.2}$$

Moreover $\widehat{\psi_t}(m) = \mathrm{Re}\,\rho_m$ and $\widehat{\phi_t}(m) = i\,\mathrm{Im}\,\rho_m$ for all $m \in \mathbb{Z}$, where

$$\rho_m = \lambda_m^2 - \omega\,\lambda_{m-t}^2 - \overline{\omega}\,\lambda_{m+t}^2. \tag{3.3}$$

Proof. This is [1, (36)–(40)]; the argument uses only $F \ge -K$, $s_0 = 0$ and $s_{-m} = s_m$, so it applies verbatim. We recall it. Since $0 \le 1+\sin(2\pi t x) \le 2$ and $F + K \ge 0$, we have $(1+\sin(2\pi t x))F \ge -(1+\sin(2\pi t x))K \ge -2K$, so $f_t \ge -4K$. Also $\lambda_0 = 2s_0 - is_{-t} + is_t = 0$, so $\int_{\mathbb{T}} f_t = 0$ and

$$f_t * f_t + 16K^2 = (f_t + 4K) * (f_t + 4K) \ge 0.$$

As $0 \le 1-\cos(2\pi tx+\pi/4) \le 2$, this gives $r_t \ge -32K^2$, and hence $\psi_t \pm \phi_t \ge -32K^2$, because $\psi_t(x) \pm \phi_t(x)$ equals $r_t(\pm x)$. Since f_t is real, so are r_t, ψ_t and ϕ_t, and (3.2) follows. Finally $1-\cos(2\pi tx+\pi/4) = 1 - \omega\, e(tx) - \overline{\omega}\, e(-tx)$ and $\widehat{f_t * f_t}(m) = \lambda_m^2$, which gives $\widehat{r_t}(m) = \rho_m$. Since r_t is real, $\widehat{r_t}(-m) = \overline{\rho_m}$, and therefore $\widehat{\psi_t}(m) = (\rho_m + \overline{\rho_m})/2$ and $\widehat{\phi_t}(m) = (\rho_m - \overline{\rho_m})/2$. □

By (3.1) and (3.3), ρ_m depends only on the vector

$$z(m) = (s_{m-2t}, s_{m-t}, s_m, s_{m+t}, s_{m+2t}) \in \mathbb{R}^5.$$

Precisely, $\rho_m = \rho(z(m))$, where for $z = (z_1, \dots, z_5)$ we set

$$\rho(z) = (2z_3 - iz_2 + iz_4)^2 - \omega\,(2z_2 - iz_1 + iz_3)^2 - \overline{\omega}\,(2z_4 - iz_3 + iz_5)^2.$$

Write $R(z) = \operatorname{Re}\rho(z)$ and $J(z) = -\operatorname{Im}\rho(z)$ for real z. Expanding gives

$$\begin{aligned} R(z) &= \tfrac{\sqrt2}{4}z_1^2 - \sqrt2 z_1z_2 - \tfrac{\sqrt2}{2}z_1z_3 - (1+\sqrt2)z_2^2 + \sqrt2 z_2z_3 + 2z_2z_4 \\ &\quad + \big(4+\tfrac{\sqrt2}{2}\big)z_3^2 + \sqrt2 z_3z_4 - \tfrac{\sqrt2}{2}z_3z_5 - (1+\sqrt2)z_4^2 - \sqrt2 z_4z_5 + \tfrac{\sqrt2}{4}z_5^2, \end{aligned} \tag{3.4}$$

$$\begin{aligned} J(z) &= -\tfrac{\sqrt2}{4}z_1^2 - \sqrt2 z_1z_2 + \tfrac{\sqrt2}{2}z_1z_3 + \sqrt2 z_2^2 + (4+\sqrt2)z_2z_3 \\ &\quad - (4+\sqrt2)z_3z_4 - \tfrac{\sqrt2}{2}z_3z_5 - \sqrt2 z_4^2 + \sqrt2 z_4z_5 + \tfrac{\sqrt2}{4}z_5^2. \end{aligned} \tag{3.5}$$

Note that $R(z_5, \dots, z_1) = R(z)$ and $J(z_5, \dots, z_1) = -J(z)$.

For $u \in \{0,1\}^5$ let

$$\mathcal{D}(u) = \{z \in \mathbb{R}^5 : z_k = 0 \text{ if } u_k = 0,\ z_k \in [\alpha, \beta] \text{ if } u_k = 1\}.$$

If $u(m) = (1_A(m-2t), 1_A(m-t), 1_A(m), 1_A(m+t), 1_A(m+2t))$, then $z(m) \in \mathcal{D}(u(m))$. Moreover $m \in B_t$ if and only if $(u_2, u_3, u_4) = (1,1,0)$ for $u = u(m)$.

Proposition 3.2. *For every $u \in \{0,1\}^5$ and every $z \in \mathcal{D}(u)$,*

$$R(z) \le 4.86 \qquad \text{and} \qquad |J(z)| \le 8.$$

If moreover $(u_2, u_3, u_4) = (1,1,0)$, then $J(z) \ge 5.27$. Consequently, for all $m \in \mathbb{Z}$,

$$\widehat{\psi_t}(m) \le 4.86, \qquad |\widehat{\phi_t}(m)| \le 8, \qquad -\operatorname{Im}\rho_m \ge 5.27 \quad \text{if } m \in B_t. \tag{3.6}$$

The proof is a finite computation with numbers in $\mathbb{Q}(\sqrt2)$, given in Appendix A.

Remark 3.3. On the vertices $z \in \{0,1\}^5$ these quantities are the ones computed in [1, Lemma 7.2]: there $\max R = 4 + 1/\sqrt2 \approx 4.707$, $\max |J| = 4 + 2\sqrt2 \approx 6.828$, and $J \ge 4 + \sqrt2 \approx 5.414$ when $(z_2, z_3, z_4) = (1,1,0)$. Proposition 3.2 states that a strict separation between the bound for R and the bound for J on B_t survives on the full boxes $\mathcal{D}(u)$. The margin is small: among the bounds obtained in Appendix A, the largest upper bound for R, which arises from the box $\mathcal{D}(0,1,1,1,0)$, is about 4.8593, and the smallest lower bound for J on B_t, which arises from $\mathcal{D}(1,1,1,0,1)$, is about 5.2766.

4. A test function and an inequality for a single translation

For $t \in \mathbb{Z}$ write $e_t(x) = e(tx)$.

Lemma 4.1. *For $t \in \mathbb{Z} \setminus \{0\}$ let*

$$g_t = F * (e_tF) - F * (e_tF) * (e_{-t}F).$$

Then $\widehat{g_t}(m) = b_m$ for all $m \in \mathbb{Z}$, where

$$b_m = s_m\, s_{m-t}\,(1 - s_{m+t}) \in \mathbb{R}.$$

The coefficients b_m satisfy the following:

(1) $b_m = 0$ *unless* $m \in B_t \cup C_t$;
(2) $\alpha^2 \le b_m \le \beta^2$ *for* $m \in B_t$;
(3) $|b_m| \le \eta$ *for* $m \in C_t$.

Moreover

$$\|g_t\|_2^2 \le \beta^4 |B_t| + \eta^2 |C_t| \qquad \text{and} \qquad \|g_t\|_1 \le 13K^3. \tag{4.1}$$

Proof. The Fourier coefficients of $e_t F$ and $e_{-t}F$ at m are s_{m-t} and s_{m+t}, so the formula for $\widehat{g_t}$ follows from the convolution rule. If $b_m \neq 0$, then $m \in A$ and $m - t \in A$, that is, $m \in B_t \cup C_t$. If $m \in B_t$, then $s_{m+t} = 0$ and $b_m = s_m s_{m-t} \in [\alpha^2, \beta^2]$. If $m \in C_t$, then $|1 - s_{m+t}| \le 1/100$ and $|b_m| \le \beta^2/100 = \eta$. Parseval's identity gives the first bound in (4.1). Since $\|e_{\pm t}F\|_1 = \|F\|_1$, Young's inequality and Lemma 2.1 give

$$\|g_t\|_1 \le \|F\|_1^2 + \|F\|_1^3 \le 4K^2 + 8K^3 \le \Big(\frac{4}{\alpha} + 8\Big) K^3 \le 13K^3. \qquad \square$$

When all coefficients equal one, g_t is the exponential sum of B_t. In general the coefficients on C_t are small but need not vanish, and they may be negative; only their squares enter below.

Proposition 4.2. *For every $t \in \mathbb{Z} \setminus \{0\}$,*

$$(5.27\,\alpha^4 - 4.86\,\beta^4)\,|B_t| \le 12.86\,\eta^2\,|C_t| + 5408\,K^8. \tag{4.2}$$

Proof. Let $G_t = g_t * g_t$, so that $\widehat{G_t}(m) = b_m^2$, and let $h_t = |g_t|$. Since the coefficients b_m are real, $g_t(-x) = \overline{g_t(x)}$; hence h_t is real and even, and its Fourier coefficients are real. Moreover $|G_t| \le h_t * h_t$ pointwise.

By Parseval's identity and Lemma 3.1,

$$i \int_{\mathbb{T}} \phi_t\, \overline{G_t} = i \sum_{m \in \mathbb{Z}} \widehat{\phi_t}(m)\, b_m^2 = \sum_{m \in \mathbb{Z}} (-\operatorname{Im} \rho_m)\, b_m^2.$$

We bound the right-hand side from below. By Lemma 4.1, only $m \in B_t \cup C_t$ contribute. For $m \in B_t$ we have $-\operatorname{Im}\rho_m \ge 5.27$ by (3.6) and $b_m^2 \ge \alpha^4$; for $m \in C_t$ we have $|\operatorname{Im}\rho_m| \le 8$ and $b_m^2 \le \eta^2$. Hence

$$\sum_{m \in \mathbb{Z}} (-\operatorname{Im}\rho_m)\, b_m^2 \ge 5.27\,\alpha^4 |B_t| - 8\,\eta^2 |C_t|.$$

For the upper bound we use (3.2), the nonnegativity of $\psi_t + 32K^2$ and of $h_t * h_t$, and Parseval's identity:

$$\Big| \int_{\mathbb{T}} \phi_t\, \overline{G_t} \Big| \le \int_{\mathbb{T}} (\psi_t + 32K^2)(h_t * h_t) = \sum_{m \in \mathbb{Z}} \widehat{\psi_t}(m)\, \widehat{h_t}(m)^2 + 32K^2 \Big(\int_{\mathbb{T}} h_t \Big)^2.$$

Since $\widehat{h_t}(m)^2 \ge 0$ and $\widehat{\psi_t}(m) \le 4.86$ for all m by (3.6), the sum is at most $4.86\,\|h_t\|_2^2 = 4.86\,\|g_t\|_2^2$. Using (4.1), we obtain

$$5.27\,\alpha^4 |B_t| - 8\,\eta^2 |C_t| \le 4.86\big(\beta^4 |B_t| + \eta^2 |C_t|\big) + 32 \cdot 13^2 K^8,$$

which is (4.2), since $32 \cdot 13^2 = 5408$. $\square$

Remark 4.3. Inequality (4.2) alone does not bound $|B_t|$, because $|C_t|$ may be much larger than $|B_t|$. For example, if $A = \{\pm 1, \dots, \pm L\}$ with $L \geq 2$ and $t = 1$, then $B_t = \{-1, L\}$, while $|C_t| = 2(L-2)$. The error terms are controlled only on average over t, by Corollary 2.4.

5. Proof of the main results

Proof of Theorem 1.2. Suppose first that $n < 3K^2$. Since $K \geq \alpha$ by Lemma 2.1 and $3 < 10^8\alpha^7$, we obtain $n < 3K^2 < 10^8 K^9$.

Suppose now that $n \geq 3K^2$. Summing (4.2) over all $t \in A$ (both positive and negative) and applying the inequality $C_{\mathrm{tot}} \leq 2B_{\mathrm{tot}}$ of Corollary 2.4, we get

$$(5.27\,\alpha^4 - 4.86\,\beta^4)\,B_{\mathrm{tot}} \leq 12.86\,\eta^2\,C_{\mathrm{tot}} + 5408\,nK^8 \leq 25.72\,\eta^2\,B_{\mathrm{tot}} + 5408\,nK^8.$$

The remaining margin is

$$M = 5.27\,\alpha^4 - 4.86\,\beta^4 - 2\cdot 12.86\,\eta^2 = \frac{58226264657}{25\cdot 10^{12}} = 0.00232905\ldots > \frac{1}{500}.$$

Hence

$$B_{\mathrm{tot}} \leq 500\cdot 5408\,nK^8 = 2{,}704{,}000\,nK^8.$$

On the other hand $B_{\mathrm{tot}} \geq n^2/(6K)$ by Corollary 2.4. Therefore

$$n \leq 6\cdot 2{,}704{,}000\,K^9 = 16{,}224{,}000\,K^9 < 10^8 K^9. \qquad \square$$

Proof of Theorem 1.1. Let $S = \{a_1, \dots, a_N\}$ and $A = S \cup (-S)$, a symmetric subset of $\mathbb{Z}\setminus\{0\}$ with $n = |A| = 2N$. Put $s_{\pm a_j} = s_j$. Then

$$F(x) = \sum_{a\in A} s_a\, e(ax) = 2\sum_{j=1}^{N} s_j \cos(2\pi a_j x),$$

so $K = 2K_{\cos}$, where $K_{\cos} = -\min_x \sum_j s_j \cos(2\pi a_j x)$. Theorem 1.2 gives $2N \leq 10^8(2K_{\cos})^9$, that is,

$$N \leq 2^8\cdot 10^8\, K_{\cos}^9 \leq 20^9 K_{\cos}^9,$$

because $2^8\cdot 10^8 = 2.56\cdot 10^{10} < 5.12\cdot 10^{11} = 20^9$. Hence $K_{\cos} \geq N^{1/9}/20$. $\square$

6. Remarks

6.1. **Upper bounds.** Since unit coefficients lie in I, every upper bound for $K(N)$ is an upper bound for $K_I(N)$. For completeness we record the Sidon construction mentioned in [1, Section 1] in the simplest case.

Proposition 6.1. *For every $m \geq 2$ there is a set S of $N = m(m-1)/2$ positive integers with*

$$\sum_{a\in S} \cos(2\pi a x) \geq -\frac{m}{2} \qquad (x \in \mathbb{R}).$$

In particular $K_I(N) \leq m/2 \leq \sqrt{N/2} + 1/2$ for these N.

Proof. Let $D = \{2^0, 2^1, \dots, 2^{m-1}\}$ and let S be the set of positive differences of elements of D. The differences $2^i - 2^j = 2^j(2^{i-j} - 1)$ with $i > j$ are pairwise distinct, since j is the 2-adic valuation of $2^i - 2^j$ and $i - j$ is then determined. Hence $|S| = m(m-1)/2$ and

$$\Big|\sum_{b\in D} e(bx)\Big|^2 = m + 2\sum_{a\in S}\cos(2\pi a x),$$

which gives the inequality. Finally $m = (1+\sqrt{1+8N})/2 \le 1+\sqrt{2N}$. □

Together with Theorem 1.1, this gives $N^{1/9}/20 \le K_I(N) \le \sqrt{N/2}+1/2$ along $N = m(m-1)/2$. For general N one obtains $K_I(N) \ll \sqrt{N}$ by removing at most $m-2$ elements from such a set.

6.2. **Scaling and constants.** If all coefficients lie in $[0.99\lambda, 1.01\lambda]$ for some $\lambda > 0$, then dividing by λ and applying Theorem 1.1 shows that the minimum is at most $-\lambda N^{1/9}/20$.

No attempt has been made to optimize the constants. The proof gives $n \le 16{,}224{,}000\,K^9$ in Theorem 1.2 (also in the case $n < 3K^2$, since $3 < 16{,}224{,}000\,\alpha^7$), and therefore the constant $1/20$ in Theorem 1.1 may be replaced by $1/12$, because $2^8 \cdot 16{,}224{,}000 \le 12^9$.

Appendix A. Proof of Proposition 3.2

Throughout this appendix $\varepsilon = 1/100$, so that $\alpha = 1-\varepsilon$ and $\beta = 1+\varepsilon$. For a real quadratic form Q on $\mathbb{R}^5$ and $u \in \{0,1\}^5$ let

$$G_Q(u) = \sum_{k:\,u_k=1} \bigl|\partial_k Q(u)\bigr|,$$

and let $H_Q(u)$ be the sum of the absolute values of the coefficients of the quadratic form obtained from Q by setting $z_k = 0$ for every k with $u_k = 0$.

Lemma A.1. *For every quadratic form Q on $\mathbb{R}^5$, every $u \in \{0,1\}^5$ and every $z \in \mathcal{D}(u)$,*

$$\bigl|Q(z) - Q(u)\bigr| \le \varepsilon\, G_Q(u) + \varepsilon^2 H_Q(u).$$

Proof. Write $z = u + h$. Then $h_k = 0$ if $u_k = 0$, and $|h_k| \le \varepsilon$ if $u_k = 1$. Since Q is a homogeneous quadratic polynomial,

$$Q(u+h) = Q(u) + \sum_{k=1}^{5} \partial_k Q(u)\, h_k + Q(h)$$

exactly. The linear term is at most $\varepsilon G_Q(u)$ in modulus. The form $Q(h)$ involves only variables h_k with $u_k = 1$, and each of its monomials is at most ε^2 in modulus, so $|Q(h)| \le \varepsilon^2 H_Q(u)$. □

Table A.1 lists, for each $u \in \{0,1\}^5$, the values $Q(u)$, $G_Q(u)$ and $H_Q(u)$ for $Q = R$ and $Q = J$, computed from (3.4) and (3.5). The entries for the reversed vector $(u_5, \dots, u_1)$ agree with those for u, except that $J(u)$ changes sign; this is consistent with the symmetries noted after (3.5). By Lemma A.1, on $\mathcal{D}(u)$ we have

$$R(z) \le R(u) + \varepsilon G_R(u) + \varepsilon^2 H_R(u), \qquad \bigl|J(z) - J(u)\bigr| \le \varepsilon G_J(u) + \varepsilon^2 H_J(u). \tag{A.1}$$

For each central triple (u_2, u_3, u_4), Table A.2 gives the largest of the four upper bounds for R in (A.1), taken over the outer bits (u_1, u_5), together with the outer bits at which it is attained; the numerical values of these bounds are rounded up to five decimal places. The last column gives the largest of the four numbers $|J(u)| + \varepsilon G_J(u) + \varepsilon^2 H_J(u)$, rounded up to three decimal places.

TABLE A.1. Center values, gradient sums and coefficient sums for R and J. The vector u is written as the string $u_1u_2u_3u_4u_5$; u_3 corresponds to the frequency m.

u	$R(u)$	$G_R(u)$	$H_R(u)$	$J(u)$	$G_J(u)$	$H_J(u)$
00000	0	0	0	0	0	0
00001	$\frac{1}{4}\sqrt{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{4}\sqrt{2}$	$\frac{1}{4}\sqrt{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{4}\sqrt{2}$
00010	$-1-\sqrt{2}$	$2+2\sqrt{2}$	$1+\sqrt{2}$	$-\sqrt{2}$	$2\sqrt{2}$	$\sqrt{2}$
00011	$-1-\frac{7}{4}\sqrt{2}$	$2+\frac{7}{2}\sqrt{2}$	$1+\frac{9}{4}\sqrt{2}$	$\frac{1}{4}\sqrt{2}$	$\frac{5}{2}\sqrt{2}$	$\frac{9}{4}\sqrt{2}$
00100	$4+\frac{1}{2}\sqrt{2}$	$8+\sqrt{2}$	$4+\frac{1}{2}\sqrt{2}$	0	0	0
00101	$4+\frac{1}{4}\sqrt{2}$	$8+\frac{1}{2}\sqrt{2}$	$4+\frac{5}{4}\sqrt{2}$	$-\frac{1}{4}\sqrt{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{3}{4}\sqrt{2}$
00110	$3+\frac{1}{2}\sqrt{2}$	$10+3\sqrt{2}$	$5+\frac{5}{2}\sqrt{2}$	$-4-2\sqrt{2}$	$8+4\sqrt{2}$	$4+2\sqrt{2}$
00111	$3-\frac{3}{4}\sqrt{2}$	$10+\frac{9}{2}\sqrt{2}$	$5+\frac{17}{4}\sqrt{2}$	$-4-\frac{5}{4}\sqrt{2}$	$8+\frac{9}{2}\sqrt{2}$	$4+\frac{15}{4}\sqrt{2}$
01000	$-1-\sqrt{2}$	$2+2\sqrt{2}$	$1+\sqrt{2}$	$\sqrt{2}$	$2\sqrt{2}$	$\sqrt{2}$
01001	$-1-\frac{3}{4}\sqrt{2}$	$2+\frac{5}{2}\sqrt{2}$	$1+\frac{5}{4}\sqrt{2}$	$\frac{5}{4}\sqrt{2}$	$\frac{5}{2}\sqrt{2}$	$\frac{5}{4}\sqrt{2}$
01010	$-2\sqrt{2}$	$4\sqrt{2}$	$4+2\sqrt{2}$	0	$4\sqrt{2}$	$2\sqrt{2}$
01011	$-\frac{11}{4}\sqrt{2}$	$\frac{11}{2}\sqrt{2}$	$4+\frac{13}{4}\sqrt{2}$	$\frac{5}{4}\sqrt{2}$	$\frac{9}{2}\sqrt{2}$	$\frac{13}{4}\sqrt{2}$
01100	$3+\frac{1}{2}\sqrt{2}$	$10+3\sqrt{2}$	$5+\frac{5}{2}\sqrt{2}$	$4+2\sqrt{2}$	$8+4\sqrt{2}$	$4+2\sqrt{2}$
01101	$3+\frac{1}{4}\sqrt{2}$	$10+\frac{5}{2}\sqrt{2}$	$5+\frac{13}{4}\sqrt{2}$	$4+\frac{7}{4}\sqrt{2}$	$8+\frac{7}{2}\sqrt{2}$	$4+\frac{11}{4}\sqrt{2}$
01110	$4+\frac{1}{2}\sqrt{2}$	$8+5\sqrt{2}$	$8+\frac{9}{2}\sqrt{2}$	0	$8+6\sqrt{2}$	$8+4\sqrt{2}$
01111	$4-\frac{3}{4}\sqrt{2}$	$8+\frac{13}{2}\sqrt{2}$	$8+\frac{25}{4}\sqrt{2}$	$\frac{3}{4}\sqrt{2}$	$8+\frac{13}{2}\sqrt{2}$	$8+\frac{23}{4}\sqrt{2}$
10000	$\frac{1}{4}\sqrt{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{4}\sqrt{2}$	$-\frac{1}{4}\sqrt{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{4}\sqrt{2}$
10001	$\frac{1}{2}\sqrt{2}$	$\sqrt{2}$	$\frac{1}{2}\sqrt{2}$	0	$\sqrt{2}$	$\frac{1}{2}\sqrt{2}$
10010	$-1-\frac{3}{4}\sqrt{2}$	$2+\frac{5}{2}\sqrt{2}$	$1+\frac{5}{4}\sqrt{2}$	$-\frac{5}{4}\sqrt{2}$	$\frac{5}{2}\sqrt{2}$	$\frac{5}{4}\sqrt{2}$
10011	$-1-\frac{3}{2}\sqrt{2}$	$2+4\sqrt{2}$	$1+\frac{5}{2}\sqrt{2}$	0	$3\sqrt{2}$	$\frac{5}{2}\sqrt{2}$
10100	$4+\frac{1}{4}\sqrt{2}$	$8+\frac{1}{2}\sqrt{2}$	$4+\frac{5}{4}\sqrt{2}$	$\frac{1}{4}\sqrt{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{3}{4}\sqrt{2}$
10101	4	8	$4+2\sqrt{2}$	0	0	$\frac{3}{2}\sqrt{2}$
10110	$3+\frac{1}{4}\sqrt{2}$	$10+\frac{5}{2}\sqrt{2}$	$5+\frac{13}{4}\sqrt{2}$	$-4-\frac{7}{4}\sqrt{2}$	$8+\frac{7}{2}\sqrt{2}$	$4+\frac{11}{4}\sqrt{2}$
10111	$3-\sqrt{2}$	$10+4\sqrt{2}$	$5+5\sqrt{2}$	$-4-\sqrt{2}$	$8+4\sqrt{2}$	$4+\frac{9}{2}\sqrt{2}$
11000	$-1-\frac{7}{4}\sqrt{2}$	$2+\frac{7}{2}\sqrt{2}$	$1+\frac{9}{4}\sqrt{2}$	$-\frac{1}{4}\sqrt{2}$	$\frac{5}{2}\sqrt{2}$	$\frac{9}{4}\sqrt{2}$
11001	$-1-\frac{3}{2}\sqrt{2}$	$2+4\sqrt{2}$	$1+\frac{5}{2}\sqrt{2}$	0	$3\sqrt{2}$	$\frac{5}{2}\sqrt{2}$
11010	$-\frac{11}{4}\sqrt{2}$	$\frac{11}{2}\sqrt{2}$	$4+\frac{13}{4}\sqrt{2}$	$-\frac{5}{4}\sqrt{2}$	$\frac{9}{2}\sqrt{2}$	$\frac{13}{4}\sqrt{2}$
11011	$-\frac{7}{2}\sqrt{2}$	$7\sqrt{2}$	$4+\frac{9}{2}\sqrt{2}$	0	$5\sqrt{2}$	$\frac{9}{2}\sqrt{2}$
11100	$3-\frac{3}{4}\sqrt{2}$	$10+\frac{9}{2}\sqrt{2}$	$5+\frac{17}{4}\sqrt{2}$	$4+\frac{5}{4}\sqrt{2}$	$8+\frac{9}{2}\sqrt{2}$	$4+\frac{15}{4}\sqrt{2}$
11101	$3-\sqrt{2}$	$10+4\sqrt{2}$	$5+5\sqrt{2}$	$4+\sqrt{2}$	$8+4\sqrt{2}$	$4+\frac{9}{2}\sqrt{2}$
11110	$4-\frac{3}{4}\sqrt{2}$	$8+\frac{13}{2}\sqrt{2}$	$8+\frac{25}{4}\sqrt{2}$	$-\frac{3}{4}\sqrt{2}$	$8+\frac{13}{2}\sqrt{2}$	$8+\frac{23}{4}\sqrt{2}$
11111	$4-2\sqrt{2}$	$8+8\sqrt{2}$	$8+8\sqrt{2}$	0	$8+6\sqrt{2}$	$8+\frac{15}{2}\sqrt{2}$

TABLE A.2. Upper bounds for R and $|J|$, grouped by the central triple.

(u_2,u_3,u_4)	(u_1,u_5)	upper bound for R	numerical value	bound for $\lvert J\rvert$
$(0,0,0)$	$(1,1)$	$\frac{10201}{20000}\sqrt{2}$	0.72132	0.361
$(0,0,1)$	$(1,0)$	$-\frac{9799}{10000}-\frac{5799}{8000}\sqrt{2}$	−2.00502	1.804
$(0,1,0)$	$(0,0)$	$\frac{10201}{2500}+\frac{10201}{20000}\sqrt{2}$	4.80172	0.361
$(0,1,1)$	$(0,0)$	$\frac{6201}{2000}+\frac{2121}{4000}\sqrt{2}$	3.85039	6.966
$(1,0,0)$	$(0,1)$	$-\frac{9799}{10000}-\frac{5799}{8000}\sqrt{2}$	−2.00502	1.804
$(1,0,1)$	$(0,0)$	$\frac{1}{2500}-\frac{9799}{5000}\sqrt{2}$	−2.77117	1.832
$(1,1,0)$	$(0,0)$	$\frac{6201}{2000}+\frac{2121}{4000}\sqrt{2}$	3.85039	6.966
$(1,1,1)$	$(0,0)$	$\frac{5101}{1250}+\frac{11009}{20000}\sqrt{2}$	4.85926	1.235

For the central triple $(1,1,0)$, which corresponds to $m \in B_t$, Table A.3 lists the lower bounds $J(u)-\varepsilon G_J(u)-\varepsilon^2 H_J(u)$ for the four choices of (u_1,u_5); their numerical values are rounded down to five decimal places.

TABLE A.3. Lower bounds for J on the boxes $\mathcal{D}(u_1,1,1,0,u_5)$.

(u_1,u_5)	lower bound for J	numerical value
$(0,0)$	$\frac{9799}{2500}+\frac{9799}{5000}\sqrt{2}$	6.69117
$(0,1)$	$\frac{9799}{2500}+\frac{68589}{40000}\sqrt{2}$	6.34458
$(1,0)$	$\frac{9799}{2500}+\frac{9637}{8000}\sqrt{2}$	5.62319
$(1,1)$	$\frac{9799}{2500}+\frac{19191}{20000}\sqrt{2}$	5.27660

Proof of Proposition 3.2. By (A.1) and Table A.2, $R(z)$ is at most

$$\frac{5101}{1250}+\frac{11009}{20000}\sqrt{2}$$

on every box $\mathcal{D}(u)$. This is less than 4.86: indeed $4.86-5101/1250=974/1250$, and $11009\sqrt{2}<15584=20000\cdot 974/1250$ because $2\cdot 11009^2=242{,}396{,}162<242{,}861{,}056=15584^2$. The last column of Table A.2 gives $|J(z)|\le 6.966<8$ on every box. By (A.1) and Table A.3, on each box $\mathcal{D}(u_1,1,1,0,u_5)$ we have

$$J(z)\ge \frac{9799}{2500}+\frac{19191}{20000}\sqrt{2},$$

and this exceeds 5.27: indeed $5.27-9799/2500=3376/2500$, and $19191\sqrt{2}>27008=20000\cdot 3376/2500$ because $2\cdot 19191^2=736{,}588{,}962>729{,}432{,}064=27008^2$.

For the consequences, recall from Section 3 that $z(m)\in\mathcal{D}(u(m))$, that $\widehat{\psi_t}(m)=R(z(m))$ and $|\widehat{\phi_t}(m)|=|J(z(m))|$, that $-\operatorname{Im}\rho_m=J(z(m))$, and that $m\in B_t$ exactly when the central triple of $u(m)$ is $(1,1,0)$. □

The supplementary material of this paper consists of three files: the program `verify.py`, its output `certificate.json`, and `README.md`. The program reconstructs (3.4) and (3.5) symbolically from the definition of ρ, computes the data of Table A.1, and checks the bounds of this appendix and the constants of Section 5 in exact arithmetic over $\mathbb{Q}(\sqrt{2})$. The output records these data for all 32 boxes,

and `README.md` describes how to run the program. The printed tables suffice for the proof.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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Email address: `math@knowledgesense.jp`